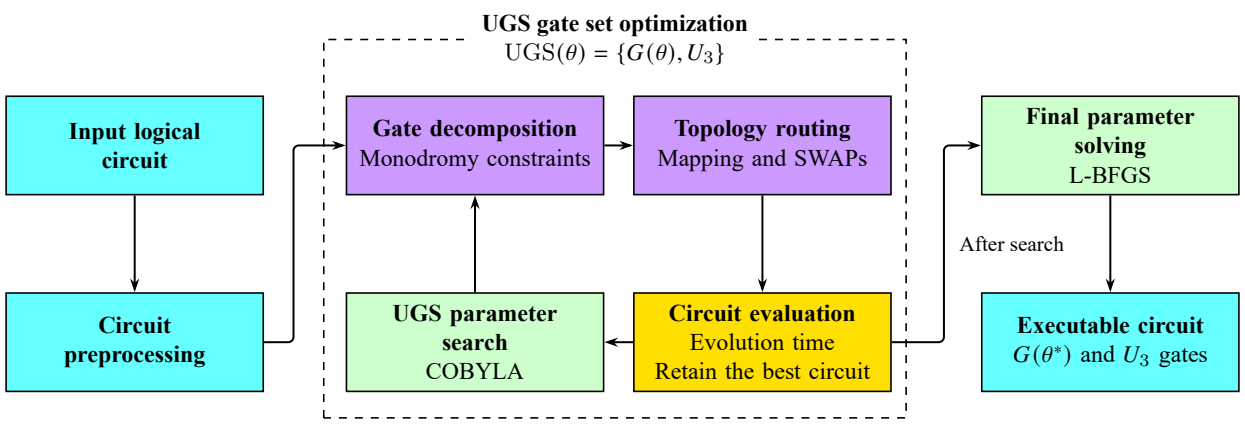

UGS gate set optimization
UGS(θ) = {G(θ), U3}
Input logical circuit
Circuit preprocessing
Gate decomposition
Monodromy constraints
Topology routing
Mapping and SWAPs
UGS parameter search
COBYLA
Circuit evaluation
Evolution time
Retain the best circuit
After search
Final parameter solving
L-BFGS
Executable circuit
G(θ*) and U3 gates